VCD-041016 SYBSC BIOCHEMISTRY-I III-SEMESTER EXAM.- 2016-17 75-MARKS 2½-HRS. PG-3×90

NOTE: 1. All questions are compulsory.

2. Figures to right indicate marks.

3. Draw labeled diagrams wherever necessary.

Q.1.A. Answer the following questions (any four) **[8M]**

1) Who first demonstrated about enzyme? Who coined the term enzyme?
2) What is the contribution of James Sumner in the field of enzymology?
3) What is the chemical nature of an enzyme?
   Make a comment on the exception on this regard.
4) Define k_{cat}
5) What is holoenzyme?
6) What do you mean by coenzyme? Give e.g.
7) What is enzyme specificity?
8) What is transition state?

Q.1.B. Answer the following questions. (Any two) **[6M]**

1) What do you mean by K_m? Give the significance of K_m.
2) How does substrate concentration affects rate of enzyme reaction?
3) Write down Koshland's induced fit theory of enzyme substrate interaction.
4) What do you mean by competitive inhibitor?
   Give the Lineweaver-Burk plot for competitive inhibitor.

Q.1.C. Answer the following questions in brief.(any one) **[6M]**

1). Write a note on classification of enzyme according to enzyme commission.
2). Derive Michaelis–Menten equation for a single enzyme- substrate reaction.

Q.2 A Answer the following (any four) **[8M]**

1) cAMP is a.......................................
   (second messenger/ first messenger/ tertiary messenger)
2) Other name of ADH is...
3) Define steroid hormone
4) What is exocrine gland ? Give example.
5) State true or False : Thyroid gland is known as the master gland.
6) Give example of peptide hormones.
7) Thyronine is a derivative of iodine and.........................
   (tryptophan /thyroxine/ histidine)
8) Explain vasopressin in short.

Q.2 B Do as directed (Any Two) **[6M]**

1) Give the physiological role of estrogens.
2) State the mode of action of steroid hormone.
3) State functions of ethylene.
4) Write down the structure of auxin and state any three functions.

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[6M]

Q.2 C Answer briefly (Any one)

1) State the physiological role of thyroxine.
2) State the physiological role of glucocorticoids.

[8M]

Q.3 A Answer the following (any four)

1) What is pH scale ?
2) What is monoproticacid ?give one example.
3) What is pH scale
4) Define physiological buffer.
5) Define Henderson Hasselbalch equation
6) What is pKa
7) Give pK values of Glysine.
8) Define isoelectric pH.

[6M]

Q.3 B Do as directed (Any Two)

1) Explain the ionization of nonpolar amino acid with suitable example.
2) Explain bicarbonate as a physiological buffer.
3) Derive ionic product of water
4) Explain the ionization of Glycine

[6M]

Q.3 C Answer briefly (Any one)

1) Calculate the pH of a buffer solution prepared by dissolving 363 mg of Tris in 10 mL of 0.2M HCl and diluting to 100 mL with water. [Tris: mw 121 g/mol and pKa = 8.08 for the conjugate acid

2) Explain the ionization of Aspartic acid with suitable titration curve.

Q.4. 1.A. Answer in short.(any one) [2M]

1) Active site 2) Define Apoenzyme

Q.4.1.B. Fill up the blanks(any three) [3M]

1) Unit of specific activity is ________________.(Katal/ U/ µmole $min^{-1}mg^{-1}$)
2) According to E.C number 6 denotes ____________ group of enzyme. (transferase/ Lyase/ Ligase)
3) ATP, FAD are________________. (Cofactor/ Coenzyme/ Prosthetic group)
4) Higher activation energy______________ the reaction rate.(Slower/ Higher/ Moderate)
5) Substrate binds to enzyme at ______________. (catalytic site/ Binding site/ action site).
6) Hydrolysis type of reaction catalysed by _______________ enzyme. (oxidoreductase / Lyase / Hydrolases)

Q.4.2.A. Answer the following : (any 1) [2M]

1) Classify hormones based on distance travelled.
2) Classify hormones based on chemistry.

Q.4.2.B. Match the following : (any 3) **[3M]**

1.Thyroid gland	a. Hypothalamus
2. Pituitary gland	b. T3 & T4
3. Master gland	c. Estrogen
4. Ovaries	d. Oxytocin
5. Shoot growth	e. Auxin
6. Root growth	f. Cytokinine

Q. 4.3.A Answer the following : (any 1) **[2M]**

1) Give one example of acidic amino acid with structure
2) Give pI value of Glycine.

Q. 4.3.B. Fill in the blanks (any 3) **[3M]**

1) Ammonium hydroxide – ammonium chloride is the example of ______ buffer. (Basic / Acidic / Neutral)
2) pK_1 value of Valine is ______ (2.29 / 9.1 / 10.5)
3) ______ is the simplest amino acid (Proline / lysine / Glycine)
4) If the pH of Tris buffer is 12 then the H^+ ion concentration of buffer is ______ (10^{-12} / 10^{-9} / 9)
5) If the pK value of molecule is 5.5 , at ______ pH it will have a zero charge. (5.5 / 10 / 9)
6) Conjugate base of NH_4OH is ______ (NH_4Cl / NaOH / CH_3COONa)

—— The End ——